📊 PhonePe Transaction Data Insights (SQL Proficiency, Data Visualization also Using Streamlit App to present the analysis results)

📌 Project Overview This project focuses on analyzing PhonePe transaction data using SQL. The dataset includes structured information across three major table categories:

🔹 Aggregated Tables 🔹 Map Tables 🔹 Top Tables The goal is to extract meaningful insights related to transaction volume, user growth, and payment behavior.

☑ Data Extraction: Collected and processed structured transaction data from CSV files, ensuring proper formatting and integration into a MySQL database for efficient querying.

☑ SQL Proficiency

📁 Table Categories 1 Aggregated Tables

Contains summarized transaction data: Used for high-level insights such as total transactions and total amount Example metrics: Total transaction count Total transaction amount.

2 Map Tables

Provides geographical insights: Data is categorized by states and districts Useful for: Regional transaction analysis User distribution mapping.

3 Top Tables

Highlights top-performing entities: Includes rankings based on: Highest transaction count Highest transaction amount Most registered users

☑ Exploratory Data Analysis (EDA) Cleaned and transformed raw data using Python (Pandas, NumPy) Conducted univariate, bivariate, and correlation analysis to understand patterns. Utilize Python libraries (e.g., Pandas, Matplotlib, Seaborn) to analyze the results from the SQL queries.

☑ Data Visualization Created insightful visualizations using Matplotlib & Seaborn Highlighted trends like transaction distribution, brand dominance, and state-wise performance.

☑ Interactive Dashboard (Streamlit) Developed a dynamic dashboard for real-time data exploration Implemented filters, KPIs, and charts for better user experience.

🔹 Analytical Thinking: Identified key insights such as regional variations in digital payment adoption, correlation between transaction count and amount, and dominant players in the market. Translated raw data into actionable business insights.

Overall, this project demonstrates the ability to transform raw transactional data into data-driven insights and interactive dashboards, showcasing strong skills in SQL, Python, and data analysis.

📌 Key Insights UPI-based transactions dominate across all regions Certain states contribute significantly to total transaction volume Rapid growth observed in registered users over time Top tables reveal concentration of transactions in major regions.

The objective is to leverage transaction data insights to increase digital payment adoption, improve user engagement, and maximize transaction value across regions and payment instruments.

Focus on High-Growth Payment Instruments:

Promote high-performing payment types (e.g., UPI) that show maximum paymentInstruments_count and paymentInstruments_amount Introduce cashback, rewards, and offers to further boost usage Encourage users of low-performing instruments to switch to faster digital methods.

Target Low-Performing Regions:

Identify regions (name) with low transaction count and amount Run localized marketing campaigns and awareness programs Collaborate with local merchants to improve digital infrastructure.

Optimize High-Value vs High-Volume Transactions:

For segments with: High count, low amount → Introduce upselling strategies Low count, high amount → Focus on retention and premium services Design targeted offers based on user spending behavior.

Strengthen Brand-Level Strategy:

Analyze brand, count, and percentage to identify leading payment platforms Partner with top-performing brands for co-branded campaigns Improve competitiveness against lower-performing brands through incentives.

Improve User Acquisition & Retention:

Use insights from transaction patterns to: Personalize offers Recommend preferred payment instruments Increase user base by targeting regions with growing adoption trends.

Data-Driven Decision Making:

Continuously monitor: Transaction trends Regional performance Payment instrument usage Build dashboards for real-time tracking and forecasting.

Increased transaction volume and value Higher user engagement and retention Expansion of digital payments in underperforming regions Stronger brand partnerships and ecosystem growth.